

POGRANICHNY, Russian Federation

WMO: 319150

Lat: 44.400N

Lon: 131.383E

Elev: 211

StdP: 98.81

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.0	(c) -21.9	(d) -29.5	(e) 0.3	(f) -22.0	(g) -27.9	(h) 0.3	(i) -20.1	(j) 9.0	(k) -12.7	(l) 7.9	(m) -12.8	(n) 2.4	(o) 280	(p) 0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	30.1	21.8	28.4	21.1	26.8	20.3	23.7	27.8	22.6	26.4	21.6	25.1	2.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.6	26.2	21.3	16.4	24.6	20.4	15.5	23.5	72.4	27.9	68.0	26.5	64.2	25.4	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	6.4	5.4	-26.2	33.1	2.1	1.2	-27.7	34.0	-28.9	34.7	-30.0	35.4	-31.5	36.3
			-26.4	25.2	2.0	1.3	-27.8	26.1	-29.0	26.9	-30.1	27.7	-31.5	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-13.9	-9.7	-2.4	6.4	12.9	17.6	21.1	20.7	15.1	7.3	-3.4	-11.8
	DBStd	13.00	4.17	4.74	5.09	4.15	3.54	3.14	2.66	2.61	3.32	4.29	6.09	4.53
	HDD10.0	2996	740	552	384	123	12	0	0	0	3	104	402	677
	HDD18.3	5051	998	785	642	358	171	49	6	8	104	341	651	936
	CDD10.0	1202	0	0	1	16	103	229	344	332	157	21	0	0
	CDD18.3	214	0	0	0	0	3	28	92	82	9	0	0	0
	CDH23.3	1573	0	0	0	9	98	271	623	490	78	3	0	0
	CDH26.7	377	0	0	0	1	25	67	176	101	9	0	0	0
(14) Wind	WSAvg	2.5	2.7	2.8	3.0	2.9	2.7	2.1	1.9	2.0	2.2	2.6	2.6	2.6
(15) Precipitation	PrecAvg	618	9	10	16	35	58	89	106	133	74	46	25	16
	PrecMax	876	35	39	50	110	113	215	267	312	151	162	75	52
	PrecMin	316	0	1	0	9	17	8	32	17	3	2	3	1
	PrecStd	125	10	9	12	22	27	52	57	66	41	34	18	13
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.6	6.2	15.1	24.7	30.2	30.9	32.6	31.0	27.9	23.0	15.1	3.8
		MCWB	-3.3	1.0	6.5	12.8	17.5	20.5	23.8	23.2	19.8	13.9	7.5	0.7
	2%	DB	-3.2	2.8	11.2	20.7	26.0	28.6	30.6	29.2	25.4	19.8	11.5	0.3
		MCWB	-5.6	-0.9	4.5	11.3	15.5	19.8	23.0	22.3	17.9	11.9	6.1	-2.4
	5%	DB	-5.3	0.2	8.0	17.6	23.1	26.4	28.7	27.6	23.7	17.2	8.2	-2.4
		MCWB	-7.4	-2.5	2.5	9.1	14.2	19.0	21.9	21.4	16.6	10.3	3.8	-4.6
	10%	DB	-7.5	-2.0	5.5	14.8	20.7	24.3	26.7	26.2	22.0	14.9	5.3	-4.6
		MCWB	-9.3	-4.4	1.0	7.4	12.7	18.1	21.3	20.6	15.6	9.4	1.8	-6.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	1.7	7.3	14.1	18.2	22.6	25.6	25.0	21.4	15.1	9.0	0.9
		MCDB	-0.7	5.0	14.5	22.6	27.7	28.6	30.7	29.5	25.2	20.6	13.5	3.6
	2%	WB	-5.5	-0.6	4.8	11.6	16.1	21.0	24.1	23.6	19.4	13.2	6.5	-2.3
		MCDB	-3.3	2.4	10.3	19.7	24.7	26.3	28.5	27.3	23.2	17.8	11.0	-0.1
	5%	WB	-7.4	-2.5	2.9	9.6	14.7	19.9	22.9	22.6	18.1	11.7	4.1	-4.4
		MCDB	-5.4	0.2	7.4	16.7	22.0	25.0	26.9	26.0	21.8	15.6	7.9	-2.4
	10%	WB	-9.2	-4.5	1.3	7.9	13.4	18.7	21.9	21.6	17.0	10.1	2.0	-6.6
		MCDB	-7.6	-1.9	5.1	13.9	19.4	23.4	25.7	24.7	20.5	13.9	5.1	-4.7
(35) Mean Daily Temperature Range	MDBR		9.5	10.7	10.6	12.2	11.8	10.6	9.0	9.2	12.2	12.3	10.3	9.2
	5% DB	MCDBR	11.5	13.1	14.4	17.4	17.0	14.4	12.4	11.4	14.1	15.9	13.9	11.8
		MCWBR	9.6	10.2	9.2	10.1	8.3	7.1	5.7	5.4	7.1	9.0	9.7	9.7
	5% WB	MCDBR	11.4	12.9	13.8	16.6	15.9	12.7	10.3	9.6	11.6	13.8	13.7	11.7
		MCWBR	9.6	10.3	9.3	10.1	8.2	7.0	5.6	5.2	6.6	8.5	9.9	9.8
(40) Clear-Sky Solar Irradiance	taub		0.323	0.372	0.458	0.566	0.585	0.566	0.528	0.485	0.438	0.456	0.386	0.337
	taud		2.084	1.960	1.830	1.679	1.703	1.830	1.989	2.102	2.137	1.948	2.033	2.056
	Ebn at Noon		758	779	755	707	712	729	754	773	776	677	666	688
	Edh at Noon		101	139	182	230	232	205	173	149	133	140	107	95
(44) All-Sky Solar Radiation	RadAvg		2.28	3.40	4.42	4.59	5.07	5.37	4.86	4.65	4.23	3.06	2.10	1.85
	RadStd		0.16	0.16	0.35	0.39	0.49	0.58	0.49	0.41	0.34	0.22	0.17	0.10

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page